



MEMS combustible gas sensor

(Model: GM-402B)

Instruction Manual

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Electronic Technology Co., Ltd.

statement

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Zhengzhou Weisheng Electronic Technology Co., Ltd.

GM-402B Combustible Gas Sensor

Product Description

MEMS combustible gas sensors utilize MEMS technology on Si-based substrates.

The gas-sensitive material used to fabricate the micro-hot plate has a conductivity of [missing information - likely related to air quality or conductivity] in clean air.

Lower-grade metal-oxide-semiconductor materials. When the detected gas is present in ambient air...

The conductivity of the sensor changes when the gas is present; the higher the concentration of the gas, the stronger the conductivity.

The higher the conductivity of the sensor, the better. A simple circuit can be used to convert the change in conductivity.

The gas is converted into an output signal corresponding to the gas concentration.



Sensor characteristics

This product utilizes MEMS technology, featuring a robust structure and high sensitivity to combustible gases; it is characterized by small size, low power consumption, high sensitivity, and fast response recovery.

It has advantages such as simple driving circuit, good stability, and long life.

Main applications

Suitable for detecting gas leaks in consumer electronics such as mobile phones and computers; also suitable for detecting combustible gas leaks in homes and commercial settings.

Leak detection devices, gas leak detectors, fire/safety detection systems.

Technical indicators

Table 1

Product Model			GM-402B
Product Type			MEMS combustible gas sensor
Standard package			Ceramic packaging
Detecting gas			methane, propane, etc.
Detection concentration			1ppm~10000ppm CH4
Standard electricity Road conditions	circuit voltage VC		~24V DC
	Heating voltage VH		3.3V±0.1V AC or DC
	Load resistor RL		Adjustable
Standard measurement Test conditions Lower gas sensitivity Component characteristics SEX	Heating resistor RH		80~±20~ (room temperature)
	Heating power consumption PH		~80mW
	Sensitive body resistance	RS	1K~±30K~(in 5000ppmCH4)
	Sensitivity S		R0(in air)/Rs(in 5000ppmCH4)~2
	Concentration slope ~		~0.9(R5000ppm/R1000ppmCH4)
Standard measurement Test conditions	Temperature and humidity: 20~±2~; 55%RH±5%RH		
	Standard test circuit		VH:3.3V±0.1V~ VC : 5.0V ± 0.1V

Sensor structure diagram

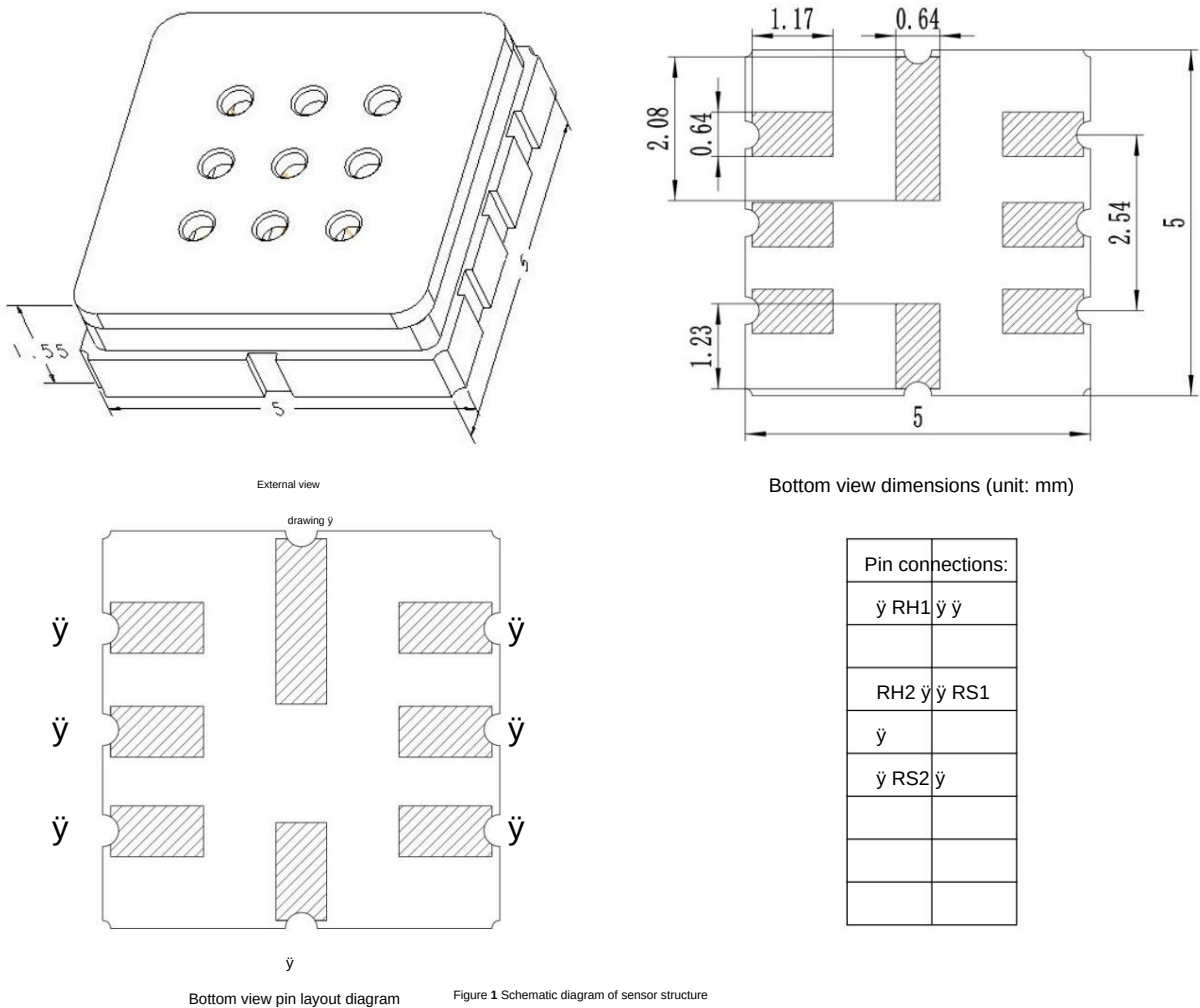


Figure 1 Schematic diagram of sensor structure

Basic circuit

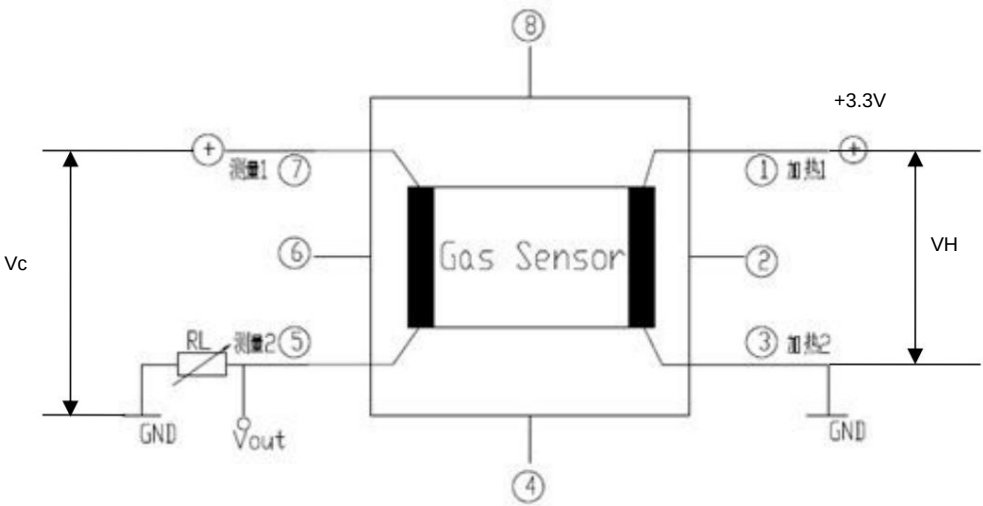


Figure 2 GM-402B test circuit

Note: The diagram above shows the basic test circuit for the GM-402B sensor. This sensor requires two voltages: the heater voltage (VH) and the test voltage (VC).

VH is used to provide a specific operating temperature for the sensor and can be supplied by either DC or AC power. Vout is the voltage across the load resistor (RL) connected in series with the sensor. VC is negative.

The on-resistor RL provides the test voltage and requires a DC power supply.

Sensor characteristic description

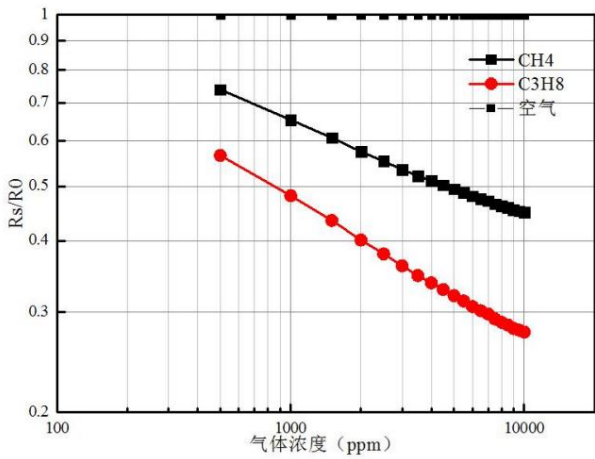


Figure 3. Typical logarithmic sensitivity characteristic curve of the sensor.

In the figure, Rs represents the resistance value of the sensor in gases of different concentrations;

R0 represents the sensor's resistance in clean air. All of the above are shown in the diagram.

All tests were conducted under standard test conditions.

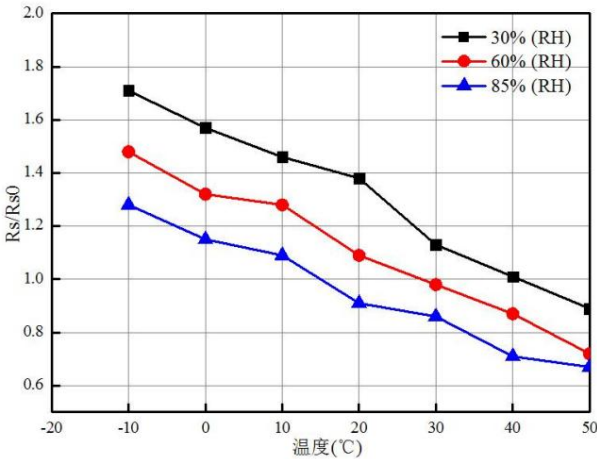


Figure 4 shows typical temperature and humidity characteristic curves of the sensor.

In the figure, Rs represents the resistance value at various temperatures and humidity levels containing

5000ppm CH4; Rs0 represents the resistance value at 20y and 55%RH containing 5000ppm CH4.

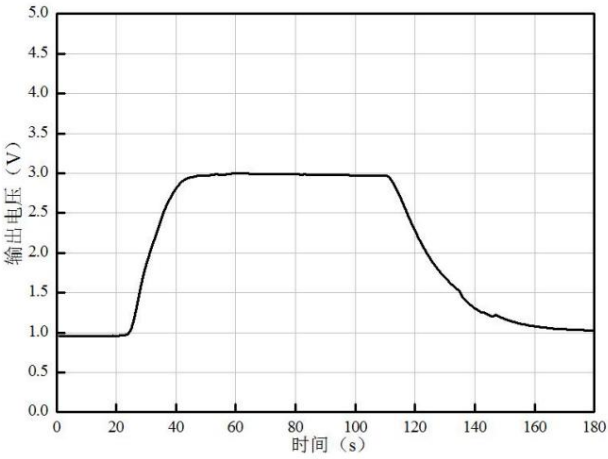


Figure 5 Response recovery curve

The output voltage in the diagram is the voltage across the load resistor (RL) connected in series with the sensor.

Voltage. The test shown in the diagram was performed under standard test conditions, using the test gas.

5000ppmCH4y

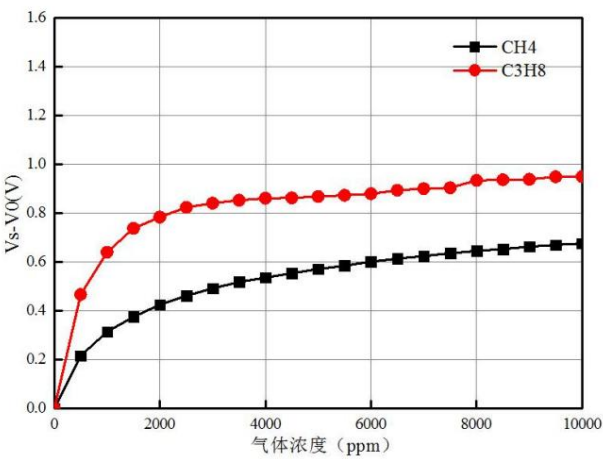


Figure 6. Linear characteristic curve of the sensor

The output voltage in the diagram is across the load resistor (RL) connected in series with the sensor.

The voltage. All tests in the diagram were performed under standard test conditions.

Long-term stability

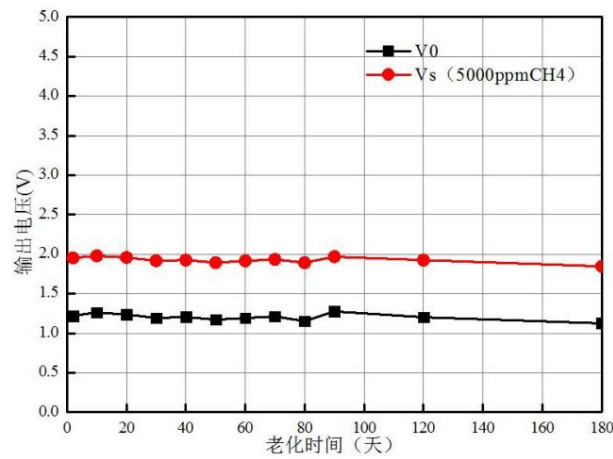


Figure 7 Long-term stable characteristic curves

Note: All tests in the figure were performed under standard test conditions. The horizontal axis represents aging time, and the vertical axis represents output voltage value.

Instructions for use:

1. Preheating time

If a sensor is stored for an extended period without power, its resistance will experience reversible drift. Therefore, the sensor needs to be preheated before use to achieve the desired internal resistance.

To maintain chemical equilibrium, the preheating voltage should be consistent with its heating voltage VH. Recommended storage times and corresponding preheating times are as follows:

Table 2

Storage time	Recommended preheating time
Less than 1 month	No less than 24 hours
1-6 months	No less than 48 hours
More than 6 months	No less than 72 hours

2. Sensor Calibration

Sensor accuracy is affected by factors such as differences in reference resistance between sensors, differences in sensitivity, temperature, humidity, interfering gases, and aging time.

The input-output relationship is nonlinear, hysteretic, and non-repeatable. Therefore, absolute concentration measurements require periodic monitoring.

Calibration (single-point/multi-point calibration can be performed across the entire measurement range) ensures accurate transmission of measurement values; relative measurements do not require calibration.

Precautions:

1. Situations that must be avoided

1.1 Exposure to volatile silicon compound vapors

The sensor should be kept away from silicone adhesives, hairspray, silicone rubber, putty, or other environments containing volatile silicone compounds. If the sensor...

The surface of the device adsorbs silicon compound vapor, and the sensor's sensitive material is coated with silicon dioxide formed from the decomposition of the silicon compound, suppressing the sensor's...

Sensitive and irreversible.

1.2 Highly corrosive environments

When sensors are exposed to high concentrations of corrosive gases (such as H₂S, SO₂, Cl₂, HCl, etc.), it can cause damage to the heating materials and the sensor itself.

Corrosion or damage to the leads can cause irreversible deterioration of the properties of sensitive materials.

1.3 Pollution from alkalis, alkali metal salts, and halogens

Sensors can also experience performance degradation after being contaminated by alkali metals, especially salt spray, or exposed to halogens such as Freon.

1.4 Contact with water

Splashing water or immersing the sensor in water can cause a decrease in its sensitivity.

1.5 Icing

Water freezing on the surface of the sensor's sensitive material can cause the sensitive layer to crack and lose its sensitive properties.

1.6 Applying voltage

Overload heating caused by overload voltage can cause irreversible damage to the sensor. Static electricity can also damage the sensor. Therefore, when connecting...

Anti-static measures should be taken when touching the sensor.

2. Situations to be avoided as much as possible

2.1 Condensate

Under indoor operating conditions, slight condensation may have a minor impact on sensor performance. If water condenses on the sensitive layer surface and remains there for a period of time...

Over time, the sensor's characteristics will degrade.

2.2 In a high concentration of gas

Regardless of whether the sensor is powered on, prolonged exposure to high concentrations of gas will affect its characteristics. For example, directly spraying lighter fluid onto the sensor...

This will cause great damage to the sensor.

2.3 Long-term exposure to extreme environments

Regardless of whether the sensor is powered on, prolonged exposure to extreme conditions, such as high humidity, high temperature, or high pollution, will affect the sensor's performance.

It has been severely affected.

2.4 Vibration

Frequent and excessive vibration can cause internal resonance in the sensor, leading to breakage. The use of pneumatic screwdrivers/ultrasonic welding during transportation and on the assembly line can also cause this.

Such vibrations are generated when the machine is in contact with the object.

2.5 Impact

If the sensor is subjected to a strong impact or collision, it may break internally.

2.6 Welding

2.6.1 Recommended conditions for reflow soldering

Neutral atmosphere;

Welding temperature 250±10°;

Avoid flux vapor.

2.6.2 Recommended conditions for manual welding

Rosin flux with the lowest chlorine content;

Welding temperature $\geq 350^{\circ}\text{C}$;

Duration $\geq 5\text{s}$.

Violating the above usage conditions will degrade the sensor's performance.

Packaging

MEMS sensors are all coated with a special protective film to prevent the effects of dust, water, atmosphere, and high temperature. This film is applied only after soldering is complete.

Remove the protective film.

We use tape and reel packaging, but other packaging methods can also be provided according to customer requirements.

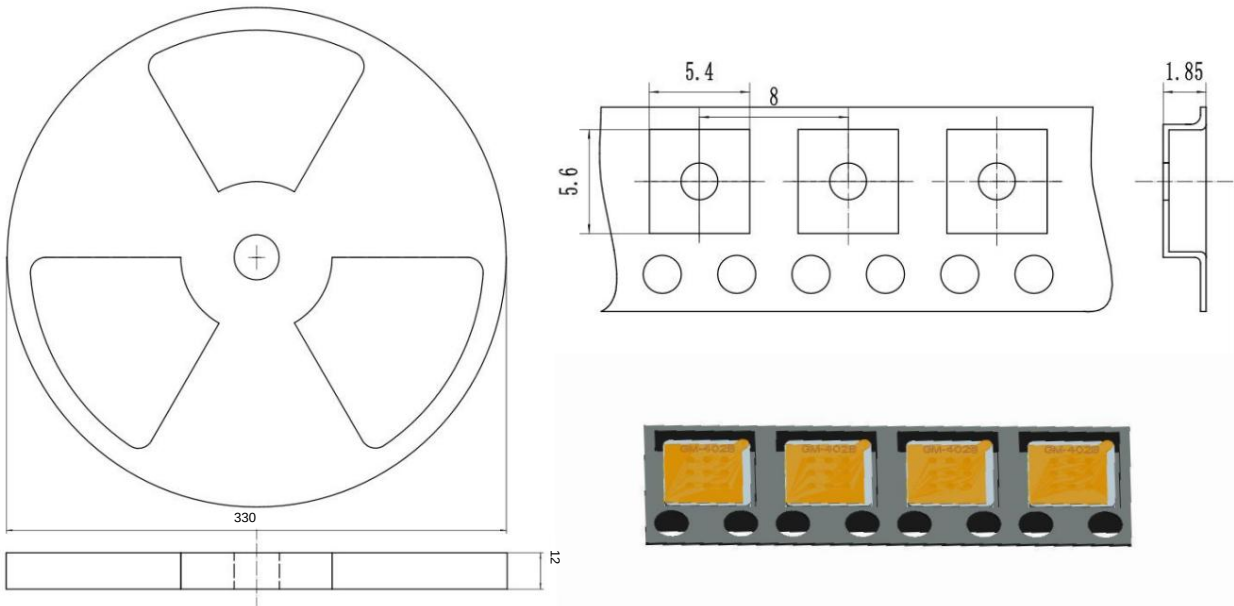


Figure 8 Sensor Packaging Diagram

Environmental Clauses

RoHS: This product complies with the current RoHS directive and does not contain any environmentally friendly substances.

Ten chemical substances restricted by environmental regulations.

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